

and

$$\mathcal{B} := \left\{ \text{MMD}_\lambda^2(p, q) \geq \sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \right)} + \hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \right\}.$$

By assumption, we have $\mathbb{P}_{p \times q \times r}(\mathcal{B}) \geq 1 - \frac{\beta}{2}$, and we want to show $\mathbb{P}_{p \times q \times r}(\mathcal{A}) \leq \beta$. Note that

$$\begin{aligned} \mathbb{P}_{p \times q \times r}(\mathcal{A} | \mathcal{B}) &= \mathbb{P}_{p \times q \times r} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \mid \mathcal{B} \right) \\ &\leq \mathbb{P}_{p \times q} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \text{MMD}_\lambda^2(p, q) - \sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \right)} \right) \\ &= \mathbb{P}_{p \times q} \left(\text{MMD}_\lambda^2(p, q) - \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \geq \sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \right)} \right) \\ &\leq \mathbb{P}_{p \times q} \left(\left| \text{MMD}_\lambda^2(p, q) - \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \right| \geq \sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \right)} \right) \\ &\leq \frac{\beta}{2} \end{aligned}$$

by Chebyshev's inequality (Chebyshev, 1899) as $\mathbb{E}_{p \times q} \left[\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \right] = \text{MMD}_\lambda^2(p, q)$. We then have

$$\begin{aligned} \mathbb{P}_{p \times q \times r}(\mathcal{A}) &= \mathbb{P}_{p \times q \times r}(\mathcal{A} | \mathcal{B}) \mathbb{P}_{p \times q \times r}(\mathcal{B}) + \mathbb{P}_{p \times q \times r}(\mathcal{A} | \mathcal{B}^c) \mathbb{P}_{p \times q \times r}(\mathcal{B}^c) \\ &\leq \frac{\beta}{2} \cdot 1 + 1 \cdot \frac{\beta}{2} \\ &= \beta. \end{aligned}$$

E.3 Proof of Proposition 3

We prove this result separately for our two MMD estimators $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ and $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ defined in Equations (3) and (6), respectively.

Variance bound for MMD estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ defined in Equation (3).

In this case, we use the fact that $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ can be written as a two-sample U -statistic as in Equation (4). As noted by Kim et al. (2022, Appendix E, Part 1) one can derive from the explicit variance formula of the two-sample U -statistic (Lee, 1990, Equation 2 p.38) that there exists some positive constant c_0 such that

$$\text{var}_{p \times q} \left(\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n) \right) \leq c_0 \left(\frac{\sigma_{\lambda, 1, 0}^2}{m} + \frac{\sigma_{\lambda, 0, 1}^2}{n} + \left(\frac{1}{m} + \frac{1}{n} \right)^2 \sigma_{\lambda, 2, 2}^2 \right)$$

for

$$\begin{aligned} \sigma_{\lambda, 1, 0}^2 &:= \text{var}_X \left(\mathbb{E}_{X', Y, Y'} [h_\lambda(X, X', Y, Y')] \right), \\ \sigma_{\lambda, 0, 1}^2 &:= \text{var}_Y \left(\mathbb{E}_{X, X', Y'} [h_\lambda(X, X', Y, Y')] \right), \\ \sigma_{\lambda, 2, 2}^2 &:= \text{var}_{X, X', Y, Y'} (h_\lambda(X, X', Y, Y')), \end{aligned}$$

where $X, X' \stackrel{\text{iid}}{\sim} p$ and $Y, Y' \stackrel{\text{iid}}{\sim} q$ are all independent of each other. Making use of Equation (20), we deduce that there exists a positive constant $c_0^\dagger$ such that

$$\text{var}_{p \times q} \left(\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n) \right) \leq c_0^\dagger \left(\frac{\sigma_{\lambda, 1, 0}^2 + \sigma_{\lambda, 0, 1}^2}{m + n} + \frac{\sigma_{\lambda, 2, 2}^2}{(m + n)^2} \right).$$

Recall that $\varphi_\lambda(u) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{u_i}{\lambda_i}\right)$ for $u \in \mathbb{R}^d$ and that $\psi := p - q$. Letting $G_\lambda = \psi * \varphi_\lambda$, we then have for all $u \in \mathbb{R}^d$

$$\begin{aligned} G_\lambda(u) &= (\psi * \varphi_\lambda)(u) \\ &= \int_{\mathbb{R}^d} \psi(u') \varphi_\lambda(u - u') du' \\ &= \int_{\mathbb{R}^d} \psi(u') k_\lambda(u, u') du' \\ &= \int_{\mathbb{R}^d} k_\lambda(u, u') (p(u') - q(u')) du' \\ &= \mathbb{E}_{X'}[k_\lambda(u, X')] - \mathbb{E}_{Y'}[k_\lambda(u, Y')]. \end{aligned}$$

Note that

$$\begin{aligned} \mathbb{E}_{X', Y'}[h_\lambda(X, X', Y, Y')] &= \mathbb{E}_{X', Y'}[k_\lambda(X, X') + k_\lambda(Y, Y') - k_\lambda(X, Y') - k_\lambda(X', Y)] \\ &= \mathbb{E}_{X'}[k_\lambda(X, X')] - \mathbb{E}_{Y'}[k_\lambda(X, Y')] \\ &\quad - (\mathbb{E}_{X'}[k_\lambda(Y, X')] - \mathbb{E}_{Y'}[k_\lambda(Y, Y')]) \\ &= G_\lambda(X) - G_\lambda(Y). \end{aligned}$$

Hence, we get

$$\begin{aligned} \sigma_{\lambda, 1, 0}^2 &:= \text{var}_X(\mathbb{E}_{X', Y, Y'}[h_\lambda(X, X', Y, Y')]) \\ &= \text{var}_X(\mathbb{E}_Y[G_\lambda(X) - G_\lambda(Y)]) \\ &= \text{var}_X(G_\lambda(X) - \mathbb{E}_Y[G_\lambda(Y)]) \\ &= \text{var}_X(G_\lambda(X)) \\ &\leq \mathbb{E}_X[G_\lambda(X)^2] \\ &= \int_{\mathbb{R}^d} G_\lambda(x)^2 p(x) dx \\ &\leq \|p\|_\infty \int_{\mathbb{R}^d} G_\lambda(x)^2 dx \\ &\leq M \|G_\lambda\|_2^2 \\ &= M \|\psi * \varphi_\lambda\|_2^2 \end{aligned}$$

and, similarly, we get

$$\sigma_{\lambda, 0, 1}^2 := \text{var}_Y(\mathbb{E}_{X, X', Y'}[h_\lambda(X, X', Y, Y')]) \leq M \|\psi * \varphi_\lambda\|_2^2.$$

For the third term, we have

$$\begin{aligned}
 \sigma_{\lambda,2,2}^2 &:= \text{var}_{X,X',Y,Y'}(h_\lambda(X, X', Y, Y')) \\
 &= \text{var}_{X,X',Y,Y'}(k_\lambda(X, X') + k_\lambda(Y, Y') - k_\lambda(X, Y') - k_\lambda(X', Y)) \\
 &\leq \mathbb{E}_{X,X',Y,Y'} \left[(k_\lambda(X, X') + k_\lambda(Y, Y') - k_\lambda(X, Y') - k_\lambda(X', Y))^2 \right] \\
 &\leq 4(\mathbb{E}_{X,X'}[k_\lambda(X, X')^2] + \mathbb{E}_{Y,Y'}[k_\lambda(Y, Y')^2] + 2\mathbb{E}_{X,Y}[k_\lambda(X, Y)^2]).
 \end{aligned}$$

Note that

$$\begin{aligned}
 \mathbb{E}_{X,Y}[k_\lambda(X, Y)^2] &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} k_\lambda(x, y)^2 p(x) q(y) \, dx dy \\
 &\leq \|p\|_\infty \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} k_\lambda(x, y)^2 \, dx \right) q(y) \, dy \\
 &= \|p\|_\infty \frac{\kappa_2}{\lambda_1 \cdots \lambda_d} \int_{\mathbb{R}^d} q(y) \, dy \\
 &\leq \frac{M\kappa_2}{\lambda_1 \cdots \lambda_d}
 \end{aligned}$$

where κ_2 depends on d and is defined in Equation (21). Similarly, we have

$$\mathbb{E}_{X,X'}[k_\lambda(X, X')^2] \leq \frac{M\kappa_2}{\lambda_1 \cdots \lambda_d} \quad \text{and} \quad \mathbb{E}_{Y,Y'}[k_\lambda(Y, Y')^2] \leq \frac{M\kappa_2}{\lambda_1 \cdots \lambda_d}. \quad (22)$$

We deduce that

$$\sigma_{\lambda,2,2}^2 := \text{var}_{X,X',Y,Y'}(h_\lambda(X, X', Y, Y')) \leq \frac{16M\kappa_2}{\lambda_1 \cdots \lambda_d}.$$

Letting $C_1(M, d) := \max \{2c_0^\dagger M, 16c_0^\dagger M\kappa_2\}$ and combining the results, we obtain

$$\begin{aligned}
 \text{var}_{p \times q}(\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n)) &\leq c_0^\dagger \left(\frac{\sigma_{\lambda,1,0}^2 + \sigma_{\lambda,0,1}^2}{m+n} + \frac{\sigma_{\lambda,2,2}^2}{(m+n)^2} \right) \\
 &\leq C_1(M, d) \left(\frac{\|\psi * \varphi_\lambda\|_2^2}{m+n} + \frac{1}{(m+n)^2 \lambda_1 \cdots \lambda_d} \right).
 \end{aligned}$$

Variance bound for MMD estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ defined in Equation (6).

The MMD estimator

$$\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n) := \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j)$$

is a one-sample U -statistic of order 2. Hence, we can apply the result of Albert et al. (2022, Lemma 10) to get

$$\text{var}_{p \times q}(\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n)) \leq \tilde{c}_0 \left(\frac{\sigma_{\lambda,1,1}^2}{n} + \frac{\sigma_{\lambda,2,2}^2}{n^2} \right)$$